



Atliq Hardwares

Consumer Goods Ad hoc Insights



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SQL Project Challenge

AGENDA

- INTRODUCTION
- ABOUT DATA
- QUERIES
- INSIGHTS

■ INTRODUCTION

Atliq Hardware, a prominent computer hardware manufacturer in India with a global customer base, aims to gain deeper insights into its product sales to support data-driven decision-making. By analyzing sales patterns, customer preferences, and market trends, the company seeks to identify growth opportunities and improve overall performance. These insights will help optimize product strategies, enhance customer satisfaction, and strengthen its competitive position in the market.

ABOUT DATA

- The dataset contains 4 fact tables: Sales Monthly, Manufacturing Cost, Pre-Invoice Deductions, and Gross Price. These tables store important measurable business data such as sales values, costs, and pricing information.
- It also includes 2 dimension tables: Customer Details and Product Details. These tables provide descriptive information that helps in understanding customer behavior and product performance more effectively.
- Atliq Hardware follows a fiscal year that starts on 1st September and ends on 31st August of the next year. This fiscal calendar is used for tracking and analyzing the company's financial and sales performance.
- The available sales data is for the fiscal years 2020 and 2021. This data helps in identifying sales trends, improving product strategies, and making better business decisions based on accurate insights.

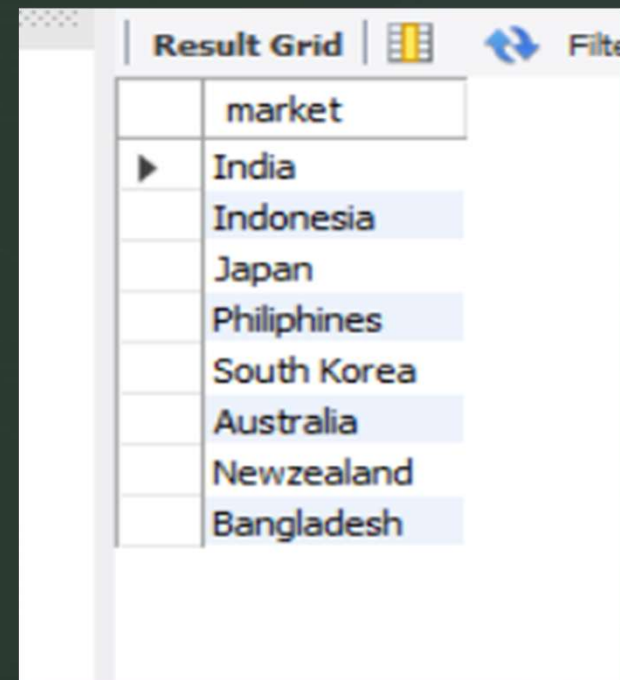
REQUEST = 1 :

Provide the list of markets in which customer "Atliq Exclusive" operates its business in the APAC region.

QUERY :

```
select
  distinct market
from dim_customer
where customer = 'Atliq Exclusive'
and region = 'APAC';
```

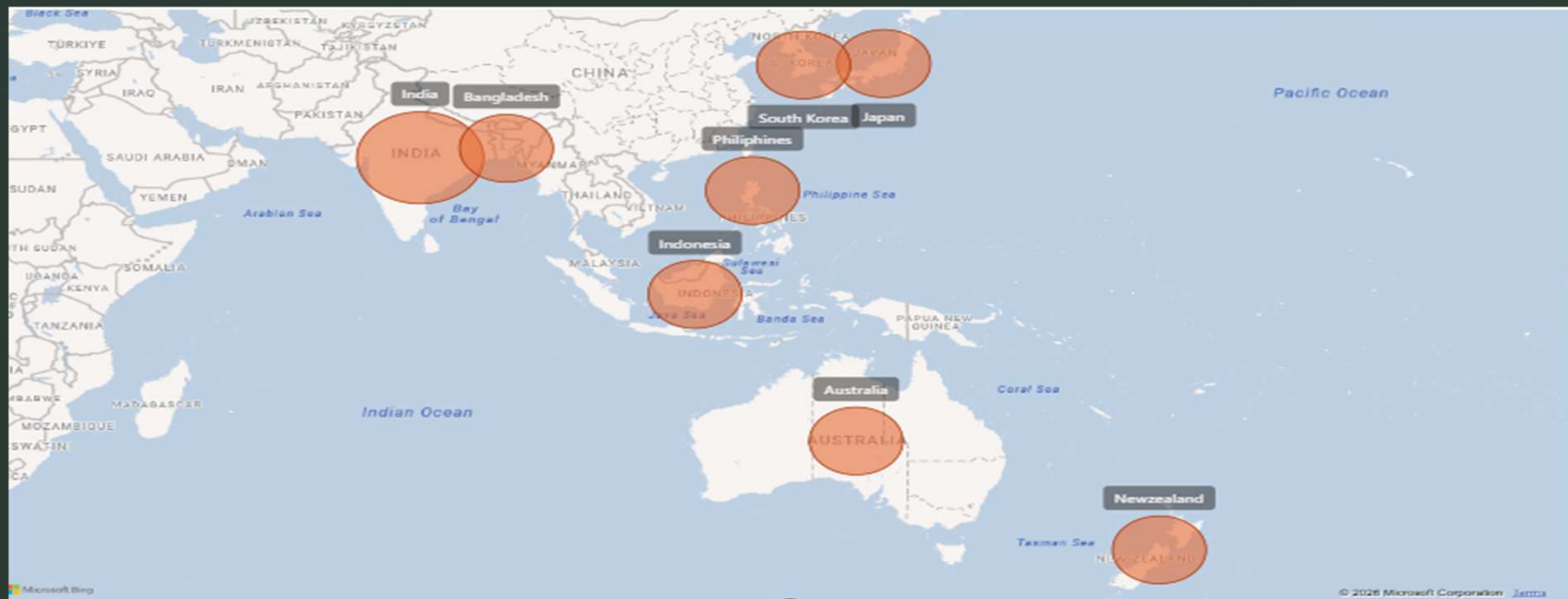
OUTPUT :



	market
▶	India
	Indonesia
	Japan
	Philipines
	South Korea
	Australia
	Newzealand
	Bangladesh

INSIGHTS :

- Atliq Exclusive has a business presence in 8 key countries across the Asia Pacific region, showing its strong expansion in international markets. The highest number of stores are located in the APAC region, while Europe has 6 stores and North America has 2 stores.
- Among these APAC countries, India generates the highest Gross Sales for Atliq Exclusive, making it the top-performing market. South Korea, Indonesia, and Australia also contribute significantly to sales, followed by the remaining countries in the region.



REQUEST: 2

What is the percentage of unique product increase in 2021 vs. 2020? -- The final output contains these fields, -- unique_products_2020 -- unique_products_2021-- percentage_chg

QUERY :

```
WITH cte AS
(
  SELECT COUNT(DISTINCT CASE WHEN fiscal_year = 2020
  THEN product_code END) AS unique_products_2020,
  COUNT(DISTINCT CASE WHEN fiscal_year = 2021 THEN product_code END) AS unique_products_2021
  FROM fact_sales_monthly)
SELECT unique_products_2020, unique_products_2021,
CONCAT(ROUND(((unique_products_2021 - unique_products_2020) * 100.0 / unique_products_2020),2),'%')
AS percentage_chg FROM cte;
```

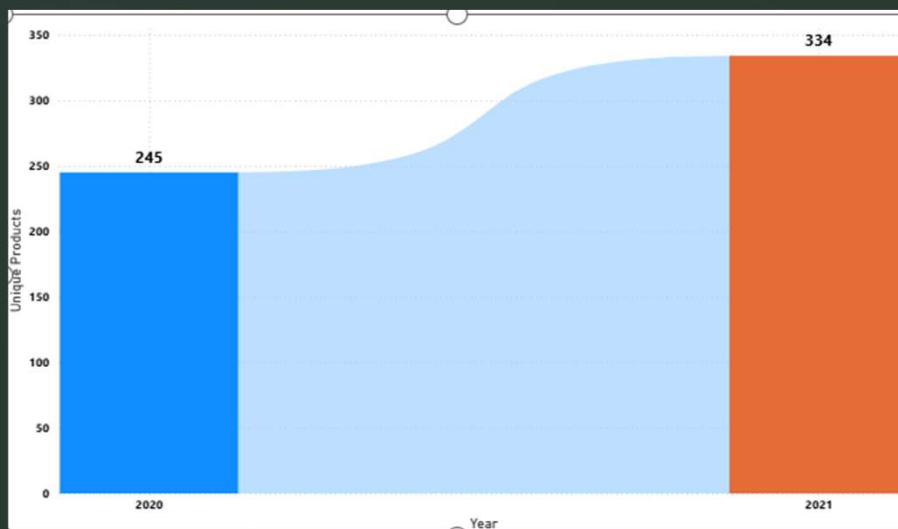
OUTPUT :

Result Grid Filter Rows: Export: Wrap Cell Content:			
	unique_products_2020	unique_products_2021	percentage_chg
▶	245	334	36.33%

INSIGHTS :

- In fiscal year 2020, Atliq Hardware had 245 unique products, which increased to 334 unique products in fiscal year 2021. This shows that the company expanded its product portfolio significantly within one year
- The percentage increase of 36.33% indicates strong business growth and a clear focus on product diversification. Adding more unique products helps the company attract more customers, improve market coverage, and strengthen its competitive position in the hardware industry.

product increase in 2021 vs. 2020



percentage_chg

36.33%

REQUEST : 3

Provide a report with all the unique product counts for each segment and sort them in descending order of product counts. -- The final output contains 2 fields, segment product_count

QUERY :

```
SELECT
    segment,
    COUNT(DISTINCT product_code) AS product_count
FROM dim_product
GROUP BY segment
ORDER BY product_count DESC;
```

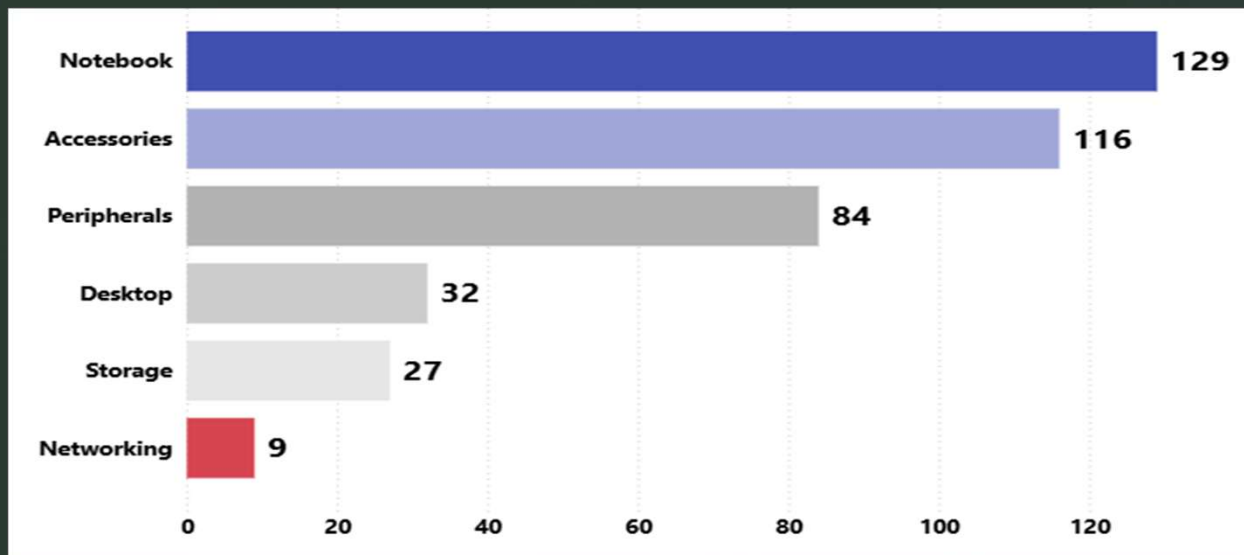
OUTPUT :

Result Grid			Filter Rows:
	segment	product_count	
▶	Notebook	129	
	Accessories	116	
	Peripherals	84	
	Desktop	32	
	Storage	27	
	Networking	9	

INSIGHTS :

- Notebook, Accessories, and Peripherals have the highest number of unique products, with more than 100 products in each segment. This shows strong demand and greater business focus on these categories
- Desktop, Storage, and Networking have fewer products, with a much smaller average product count. These segments may have lower market demand or limited business priority
- The Product Development team should focus on improving these lower-performing segments. New designs and innovation can help Atliq Hardware stay competitive and increase future growth

unique product counts for each segment



REQUEST : 4

Follow-up: Which segment had the most increase in unique products in 2021 vs 2020?

The final output contains these fields, segment product_count_2020, product_count_2021 difference

QUERY :

```
WITH segment_products AS
(SELECT p.segment,
    COUNT(DISTINCT CASE
        WHEN s.fiscal_year = 2020 THEN s.product_code
    END) AS product_count_2020,
    COUNT(DISTINCT CASE
        WHEN s.fiscal_year = 2021 THEN s.product_code
    END) AS product_count_2021
FROM fact_sales_monthly s
JOIN dim_product p
ON s.product_code = p.product_code
GROUP BY p.segment)
SELECT
    segment,
    product_count_2020,
    product_count_2021,
    (product_count_2021 - product_count_2020) AS difference
FROM segment_products
ORDER BY difference DESC;
```

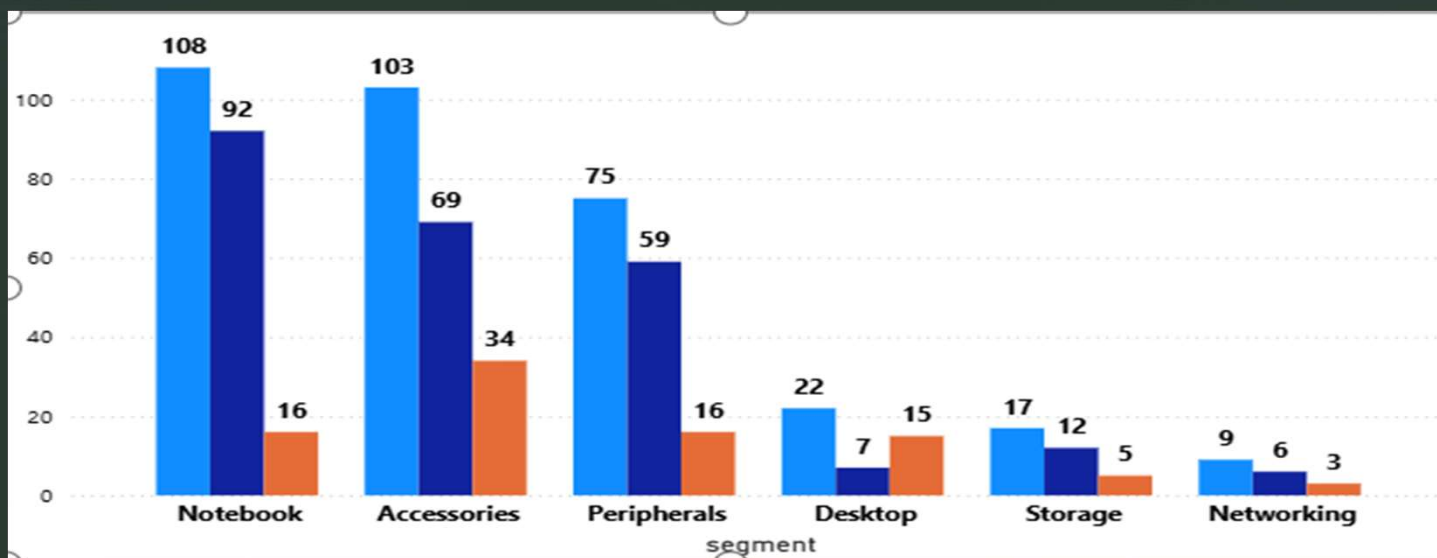
OUTPUT :

Result Grid Filter Rows: Export: Wrap Cell Content:				
	segment	product_count_2020	product_count_2021	difference
▶	Accessories	69	103	34
	Notebook	92	108	16
	Peripherals	59	75	16
	Desktop	7	22	15
	Storage	12	17	5
	Networking	6	9	3

INSIGHTS :

- Accessories had the highest increase in unique products from 2020 to 2021, growing from 69 to 103 products with a difference of 34. This shows strong business focus on this segment.
- Notebook and Peripherals also showed good growth with 16 additional products each, while Desktop increased by 15 products. These segments are important for overall sales growth
- Networking had the lowest increase with only 3 new products. This suggests that lower-growth segments may need more improvement and product development in the future

Segment-wise Unique Product Count and Growth Difference



REQUEST : 5

Get the products that have the highest and lowest manufacturing costs.-- The final output should contain these fields,-- product_code product ,manufacturing_cost

QUERY :

```
SELECT
    p.product_code,
    p.product,
    CONCAT('$', ROUND(fmc.manufacturing_cost, 2)) AS manufacturing_cost
FROM dim_product p
JOIN fact_manufacturing_cost fmc
ON p.product_code = fmc.product_code
WHERE fmc.manufacturing_cost =
(SELECT MAX(manufacturing_cost) FROM fact_manufacturing_cost)
OR fmc.manufacturing_cost =
( SELECT MIN(manufacturing_cost) FROM fact_manufacturing_cost)
ORDER BY fmc.manufacturing_cost DESC;
```

OUTPUT :

product_code	product	manufacturing_cost
A6120110206	AQ HOME Allin1 Gen 2	\$240.54
A2118150101	AQ Master wired x1 Ms	\$0.89

INSIGHTS :

- AQ HOME Allin1 Gen 2 has the highest manufacturing cost at \$240.54, making it the most expensive product to produce. This may be due to premium features and higher production requirements
- AQ Master wired x1 Ms has the lowest manufacturing cost at \$0.89, showing it is a low-cost product. This cost difference helps the company improve pricing and profitability decisions



AQ HOME Allin1 Gen 2 has the highest manufacturing cost at \$240.54



AQ Master wired x1 Ms has the lowest manufacturing cost at \$0.89

REQUEST : 6

Generate a report which contains the top 5 customers who received an average high pre_invoice_discount_pct for -- the fiscal year 2021 and in the Indian market. -- The final output contains these fields, customer_code, customer average_discount_percentage

QUERY :

```
SELECT
    c.customer_code,
    c.customer,
    ROUND(AVG(d.pre_invoice_discount_pct) * 100, 2) AS average_discount_percentage
FROM fact_pre_invoice_deductions d
JOIN dim_customer c
ON d.customer_code = c.customer_code
WHERE d.fiscal_year = 2021
AND c.market = 'India'
GROUP BY c.customer_code, c.customer
ORDER BY average_discount_percentage DESC
LIMIT 5;
```

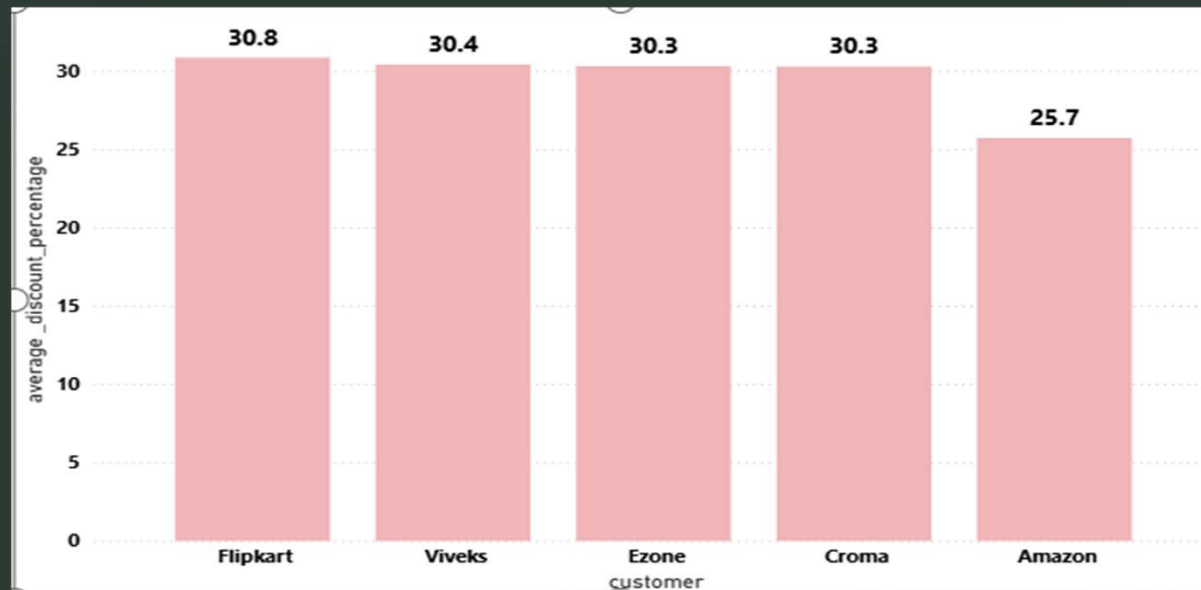
OUTPUT :

	customer_code	customer	average_discount_percentage
▶	90002009	Flipkart	30.83
	90002006	Viveks	30.38
	90002003	Ezone	30.28
	90002002	Croma	30.25
	90002016	Amazon	29.33

INSIGHTS :

- Flipkart got the highest average discount in 2021 with 30.83%, followed by Viveks and Ezone. This means these customers received more discount compared to other customers in India.
- Amazon and Croma also got good discounts but slightly less than the top 3. Giving higher discounts to big customers helps Atliq Hardware increase sales and keep strong customer relationships.

Top 5 Customers by Average Pre-Invoice Discount Percentage (FY 2021 - India)



REQUEST : 7

Get the complete report of the Gross sales amount for the customer “Atliq Exclusive” for each month . -- This analysis helps to get an idea of low and high-performing months and take strategic decisions. -- The final report contains these columns: Month Year Gross sales Amount

QUERY :

```
SELECT
    MONTHNAME(s.date) AS month_name,
    YEAR(s.date) AS year_,
    CONCAT('$', ROUND(SUM(s.sold_quantity * g.gross_price) / 1000000, 2)) AS gross_sales_amount_millions
FROM fact_sales_monthly s
JOIN fact_gross_price g
ON s.product_code = g.product_code
AND s.fiscal_year = g.fiscal_year
JOIN dim_customer c
ON s.customer_code = c.customer_code
WHERE c.customer = 'Atliq Exclusive'
GROUP BY MONTHNAME(s.date), YEAR(s.date)
ORDER BY year_;
```

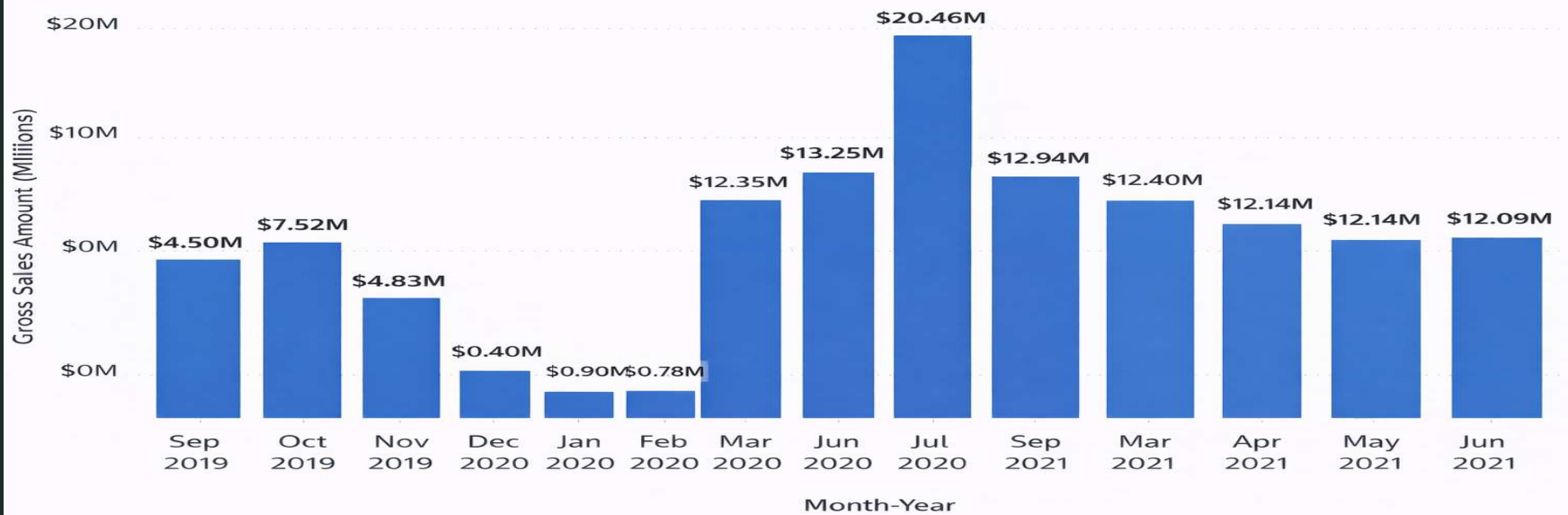
OUTPUT :

	month_name	year_	gross_sales_amount_millions
►	September	2019	\$4.50
	October	2019	\$5.14
	November	2019	\$7.52
	December	2019	\$4.83
	January	2020	\$4.74
	February	2020	\$4.00
	March	2020	\$0.38
	April	2020	\$0.40
	May	2020	\$0.78
	June	2020	\$1.70
	July	2020	\$2.55
	August	2020	\$2.79
	September	2020	\$12.35
	October	2020	\$13.22
	November	2020	\$20.46
	December	2020	\$12.94
	January	2021	\$12.40
	February	2021	\$10.13
	March	2021	\$12.14
	April	2021	\$7.31
	May	2021	\$12.15
	June	2021	\$9.82
	July	2021	\$12.09
	August	2021	\$7.18

INSIGHTS :

- Gross sales for Atliq Exclusive increased strongly from September 2020, with November 2020 reaching the highest sales at \$20.46 million. This shows strong business growth and high customer demand.
- The lowest sales were in March and April 2020, where sales dropped below \$1 million. In 2021, sales stayed stable and mostly remained above \$10 million, showing consistent performance.

Monthly Gross Sales of Atliq Exclusive



REQUEST : 8

In which quarter of 2020, got the maximum total_sold_quantity? -- The final output contains these fields sorted by the total_sold_quantity, -- Quarter total_sold_quantity

QUERY :

```
SELECT
CASE
    WHEN MONTH(date) IN (9,10,11) THEN 'Q1'
    WHEN MONTH(date) IN (12,1,2) THEN 'Q2'
    WHEN MONTH(date) IN (3,4,5) THEN 'Q3'
    WHEN MONTH(date) IN (6,7,8) THEN 'Q4'
END AS Quarter,

SUM(sold_quantity) AS total_sold_quantity

FROM fact_sales_monthly
WHERE fiscal_year = 2020
GROUP BY Quarter
ORDER BY total_sold_quantity DESC;
```

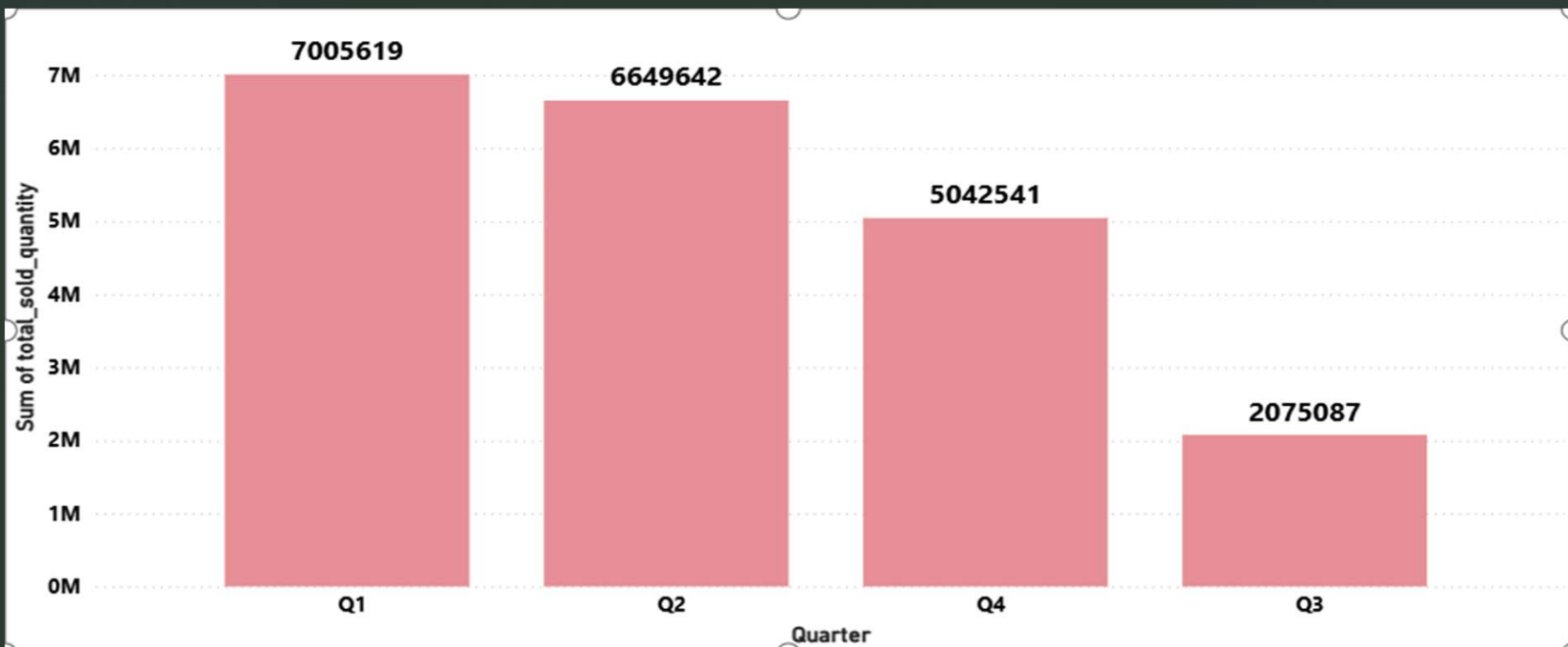
OUTPUT :

Result Grid			Filter Rows:
	Quarter	total_sold_quantity	
▶	Q1	7005619	
	Q2	6649642	
	Q4	5042541	
	Q3	2075087	

INSIGHTS :

- Q1 recorded the highest total sold quantity in 2020 with 7,005,619 units, followed closely by Q2 with 6,649,642 units
- This shows that product demand was strongest in the first half of the fiscal year, especially during September to February
- Q3 had the lowest sales with 2,075,087 units, which may indicate lower market demand or seasonal slowdown during that period

Quarter-wise Total Sold Quantity in 2020



REQUEST : 9

Which channel helped to bring more gross sales in the fiscal year 2021 and -- the percentage of contribution?-- The final output contains these fields, channel gross_sales_mln percentage

QUERY :

```
SELECT
    c.channel,
    ROUND(SUM(s.sold_quantity * g.gross_price) / 1000000, 2) AS gross_sales_mln,
    CONCAT(ROUND((SUM(s.sold_quantity * g.gross_price) * 100.0) /
        SUM(SUM(s.sold_quantity * g.gross_price)) OVER (),2),'%') AS percentage
FROM fact_sales_monthly s
JOIN fact_gross_price g
ON s.product_code = g.product_code
AND s.fiscal_year = g.fiscal_year
JOIN dim_customer c
ON s.customer_code = c.customer_code
WHERE s.fiscal_year = 2021
GROUP BY c.channel
ORDER BY gross_sales_mln DESC;
```

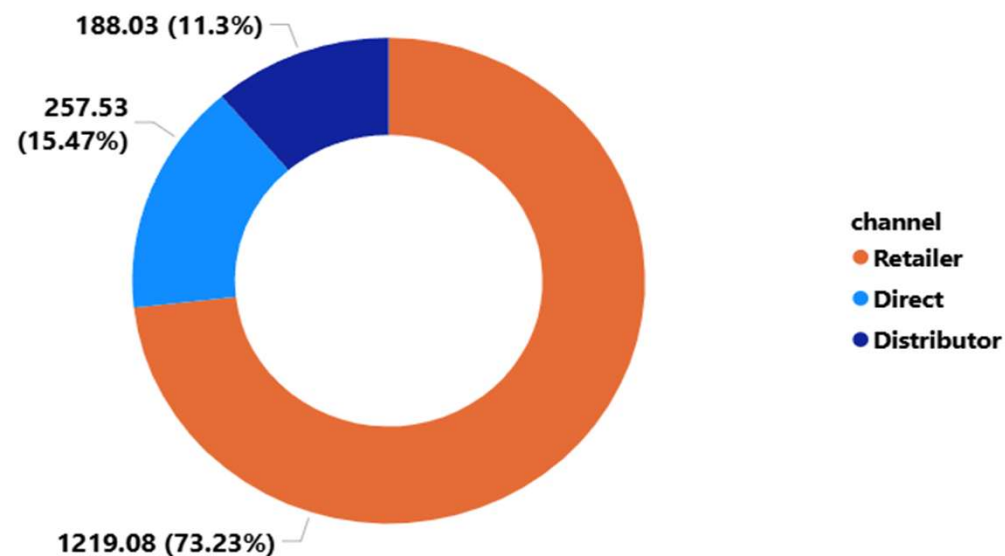
OUTPUT :

Result Grid   Filter Rows:			
	channel	gross_sales_mln	percentage
▶	Retailer	1219.08	73.23%
	Direct	257.53	15.47%
	Distributor	188.03	11.30%

INSIGHTS :

- Retailer channel contributed the highest gross sales in fiscal year 2021 with \$1219.08 million, accounting for 73.23% of total sales
- Direct channel generated \$257.53 million (15.47%), while Distributor contributed the lowest with \$188.03 million (11.30%).
- This shows that the Retailer channel is the strongest sales driver and plays a major role in overall business growth.

Channel-wise Gross Sales Contribution – FY 2021



REQUEST : 10

Get the Top 3 products in each division that have a high total_sold_quantity in the fiscal_year 2021?-- The final output contains these fields, division product_codeproduct total_sold_quantity rank_order

QUERY :

```
WITH product_sales AS( SELECT
    p.division,
    p.product_code,
    p.product,
    SUM(s.sold_quantity) AS total_sold_quantity,
    RANK() OVER
    (PARTITION BY p.division
    ORDER BY SUM(s.sold_quantity) DESC ) AS rank_order
FROM fact_sales_monthly s
JOIN dim_product p
ON s.product_code = p.product_code
WHERE s.fiscal_year = 2021
GROUP BY p.division, p.product_code, p.product)
SELECT division,
    product_code,
    product,
    total_sold_quantity,
    rank_order
FROM product_sales
WHERE rank_order <= 3;
```

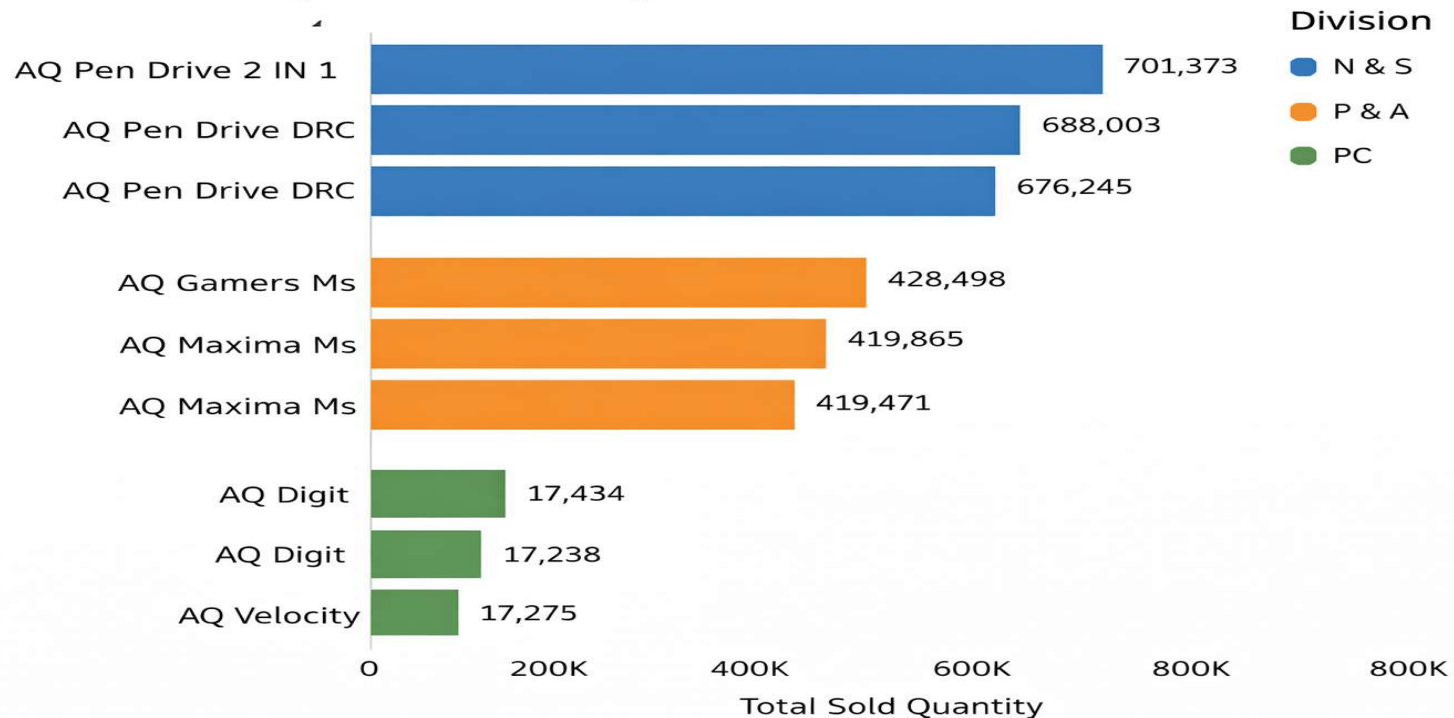
OUTPUT :

Result Grid		 Filter Rows:			Export: 	Wrap Cell Content: 
	division	product_code	product	total_sold_quantity	rank_order	
▶	N & S	A6720160103	AQ Pen Drive 2 IN 1	701373	1	
	N & S	A6818160202	AQ Pen Drive DRC	688003	2	
	N & S	A6819160203	AQ Pen Drive DRC	676245	3	
	P & A	A2319150302	AQ Gamers Ms	428498	1	
	P & A	A2520150501	AQ Maxima Ms	419865	2	
	P & A	A2520150504	AQ Maxima Ms	419471	3	
	PC	A4218110202	AQ Digit	17434	1	
	PC	A4319110306	AQ Velocity	17280	2	
	PC	A4218110208	AQ Digit	17275	3	

INSIGHTS :

- N & S division recorded the highest total sold quantity, with AQ Pen Drive 2 IN 1 leading at 701,373 units, followed by AQ Pen Drive DRC products
- P & A division also showed strong demand, where AQ Gamers Ms ranked first with 428,498 units sold
- PC division had the lowest total sold quantity, with AQ Digit and AQ Velocity selling much less compared to other divisions, showing lower product demand in this segment.

Top 3 Products by Division – FY 2021



THANK YOU

Mohd Nafees